

Explaining FTSE 100 Monthly Returns Using S&P 500 Returns

Data source: Bloomberg Terminal | Software: EViews and Microsoft Excel

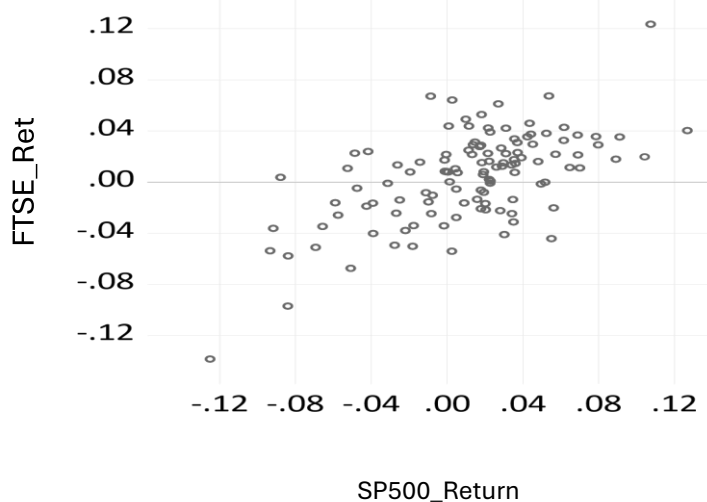
Research Question: To what extent do S&P 500 monthly returns explain FTSE 100 monthly returns?

Model: $FTSE_Return = c + \beta_1(SP500_Return) + \text{error}$

Method: Monthly return data for the FTSE 100 and S&P 500 was imported into EViews. An ordinary least squares regression was used to test whether S&P 500 monthly returns helped explain FTSE 100 monthly returns.

Regression Output:

Dependent Variable: FTSE_RETURN Method: Least Squares Date: 05/24/26 Time: 19:10 Sample (adjusted): 2016M06 2026M04 Included observations: 119 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.000592	0.002650	-0.223398	0.8236
SP500_RETURN	0.482801	0.058128	8.305794	0.0000
R-squared	0.370921	Mean dependent var		0.004916
Adjusted R-squared	0.365544	S.D. dependent var		0.035143
S.E. of regression	0.027992	Akaike info criterion		-4.297135
Sum squared resid	0.091675	Schwarz criterion		-4.250427
Log likelihood	257.6795	Hannan-Quinn criter.		-4.278168
F-statistic	68.98622	Durbin-Watson stat		2.077149
Prob(F-statistic)	0.000000			



Interpretation:

The regression results show a positive and statistically significant relationship between S&P 500 monthly returns and FTSE 100 monthly returns. The coefficient for S&P 500 returns is 0.4828, meaning that a 1 percentage point increase in S&P 500 monthly returns is associated with an approximate 0.48 percentage point increase in FTSE 100 monthly returns. The p-value is 0.0000, which suggests that the relationship is statistically significant. The R-squared value of 0.371 indicates that S&P 500 monthly returns explain around 37.1% of the variation in FTSE 100 monthly returns.

Conclusion:

Overall, the results suggest that US equity market movements are an important driver of UK equity market returns. However, because the R-squared is 37.1%, other factors also influence FTSE 100 performance, such as UK-specific market conditions, exchange rates, interest rates, sector composition, and investor sentiment.